

KATIE E. BRISTOL

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EDUCATION

Ph.D. in Geological Sciences	2024
University of Florida, Gainesville, FL Advisor: Dr. Courtney Sprain	
M.S. in Geophysics	2020
Michigan Technological University, Houghton, MI Advisor: Dr. Aleksey Smirnov	
B.S. in Applied Geophysics <i>w/ minor in Geological Engineering</i>	2018
Michigan Technological University, Houghton, MI Field Courses: University of Canterbury, New Zealand	

RESEARCH EXPERIENCE

Research Scientist , <i>Institute for Rock Magnetism, University of Minnesota</i>	2026 - present
NSF Postdoctoral Research Fellow , <i>PMag Lab, Purdue University</i>	2025 - 2026
Research Assistant , <i>Neil Opdyke Paleomagnetic Lab, University of Florida</i>	2020 - 2024
Lab Management Assistant , <i>Earth Magnetism Lab, Michigan Tech. University</i>	2017 - 2020
Research Assistant , <i>Earth Magnetism Lab, Michigan Tech. University</i>	2015 - 2020
NASA Michigan Space Grant Fellow , <i>Michigan Tech. University</i>	2018 - 2019
Summer Undergraduate Research Fellow , <i>Michigan Tech. University</i>	2016 - 2017
NSF Summer REU Intern , <i>Michigan Tech. University</i>	2015

PEER-REVIEWED PUBLICATIONS

In print:

1. **Bristol, K.E.**, Sprain, C. J., Mittal, T., Monteiro, A., Duraiswami, R., Tremblay, M. M., & Mijum, M. (2026). Geomagnetic variability in a post-superchron geodynamo: Insights from the Deccan Traps. *Journal of Geophysical Research: Solid Earth*, 131(1), e2025JB033124. <https://doi.org/10.1029/2025JB033124>
2. **Bristol, K.E.**, Sprain, C. J., Meert, J. G., Yasar, I. D., Pandit, M. K., Sinha, A. K., & Dann, A. B. (2025). Absolute palaeointensity estimates from Precambrian India and the long-term thermal evolution of the Earth. *Geophysical Journal International*, 241(1), 291–307. <https://doi.org/10.1093/gji/ggaf038>
3. Mijum, M., **Bristol, K.E.**, Bono, R. K., Sprain, C. J., Lifton, N., & Tremblay, M. M. (2025). A model framework for scaling pre-quaternary cosmogenic nuclide production rates. *Geochemistry, Geophysics, Geosystems*, 26(1), e2024GC012020. <https://doi.org/10.1029/2024GC012020>
4. Engbers, Y. A., Thallner, D., Bono, R. K., Sprain, C. J., Murray, M. J., **Bristol, K.E.**, Handford, B., Torsvik, T., & Biggin, A. J. (2024). A global paleosecular variation database for the Paleogene: Stationary secular variation behavior since the Triassic? *Geochemistry, Geophysics, Geosystems*, 25(6), e2023GC011203. <https://doi.org/10.1029/2023GC011203>

5. **Bristol, K.E.**, Smirnov, A. V., Piispa, E. J., Ramirez Navas, M. R., Kostrov, A., & Kulakov, E. V. (2023). Magnetic characterization of the Daule chondrite (Ecuador's first meteorite fall): The case of elusive tetraenaite? *Icarus*, 404, 115684. <https://doi.org/10.1016/j.icarus.2023.115684>
6. Smirnov, A. V., Kulakov, E. V., Foucher, M. S., & **Bristol, K.E.** (2017). Intrinsic paleointensity bias and the long-term history of the geodynamo. *Science Advances*, 3(2), e1602306. <https://doi.org/10.1126/sciadv.1602306>

ABSTRACTS AND PRESENTATIONS

For a complete list, see my [Google Scholar page](#).

Scientific Talks:

1. **Bristol, K.E.**, & Borlina, C. S. (2026). Early Solar System Magnetic Records in Calcium Aluminum Rich Inclusions from the ALHA77307 CO Chondrite. In *88th Annual Meeting of the Meteoritical Society* (Vol. 3162, p. 5412).
2. Borlina, C. S., **Bristol, K.E.**, Doctor, R., & Feinberg, J. M. (2026). Evidence of a Solar Nebula Magnetic Field in Samples Returned from Bennu. In *88th Annual Meeting of the Meteoritical Society* (Vol. 3162, p. 5134).
3. Borlina, C. S., **Bristol, K.E.**, Doctor, R., & Feinberg, J. M. (2026). Evidence of a Solar Nebula Magnetic Field in Samples Returned from Bennu. In *57th Lunar and Planetary Science Conference* (Vol. 3160, p. 1453).
4. Borlina, C. S., **Bristol, K.E.**, Doctor, R., & Feinberg, J. M. (2025). Potential Evidence of a Solar Nebula Magnetic Field in Samples Returned from Bennu. In *Workshop on Bennu and Ryugu: Samples from the Early Solar System* (Vol. 3094, p. 2020).
5. **Bristol, K.E.**, Borlina, C. S., & Feinberg, J. M. (2025). Magnetized in Brines: Evidence for Robust Magnetic Carriers from Bennu's Returned Samples. In *Workshop on Bennu and Ryugu: Samples from the Early Solar System* (Vol. 3094, p. 2045).
6. Sprain, C. J., **Bristol, K.E.**, Mittal, T., Fendley, I. M., Tremblay, M. M., Mijjū, M., Monteiro, A., & Duraiswami, R. (2025). Using Quantitative Paleosecular Variation Analysis to Refine the Tempo of Deccan Eruptions. In *Geological Society of America Abstracts* (Vol. 57, No. 6). <https://doi.org/10.1130/abs/2025AM-10066>
7. **Bristol, K.E.** (2024). Fast and Furious: Magnetic Drift - Using Quantitative Paleosecular Variation Techniques to Quantify the Tempo of Large Igneous Provinces. In *Magnetic NetworkZ (MagNetZ) Seminars* (Vol. 2024). <https://doi.org/10.7288/V4/ERDA/2733>
8. **Bristol, K.E.**, Sprain, C. J., Griffis, A., Mittal, T., Fendley, I. M., Duraiswami, R. A., Monteiro, A., Mijjū, M., & Tremblay, M. M. (2023). Assessing Eruptive Hiatus Durations of the Deccan Traps Large Igneous Province Using Quantitative Paleosecular Variation Analysis. In *AGU Fall Meeting Abstracts* (Abstract GP33A-01).
9. **Bristol, K.E.**, Sprain, C. J., Piispa, E. J., Smirnov, A. V., Meert, J. G., & Dann, A. (2022). New Absolute Paleointensity Estimates from Mafic Dikes of India and the Variability of the Precambrian Geomagnetic Field. In *AGU Fall Meeting Abstracts* (Abstract GP26A-02).
10. Mijjū, M., **Bristol, K.E.**, Tremblay, M. M., & Sprain, C. J. (2021). A Model Framework for Cosmogenic Nuclide Production Rates Through Deep Time. In *Geological Society of America Abstracts* (Vol. 53, No. 6, p. 364512). <https://doi.org/10.1130/abs/2021AM-364512>

11. **Bristol, K.E.**, Piispa, E. J., & Smirnov, A. V. (2020). Thermomagnetic behavior of extraterrestrial minerals: An overview. In *AGU Fall Meeting Abstracts* (Abstract GP004–05).

Posters and Presentations:

* = presenting author

1. **Bristol, K.E.**, Borlina, C. S., & Feinberg, J. M. (2025). From Bulk Rock Magnetism to Individual Inclusions: A Detailed Magnetic Characterization of Refractory Inclusions in Chondrites for Paleomagnetic Analysis. In *AGU Fall Meeting Abstracts* (Abstract P23H–2754).
2. Sprain, C. J., Holliday, M., **Bristol, K.E.**, Thallner, D., & Fu, R. R. (2025). A Framework for Leveraging Clinkers as a Full-Vector Paleomagnetic Resource in Underrepresented Terrains. In *AGU Fall Meeting Abstracts* (Abstract GP31A–0305).
3. **Bristol, K.E.**, Borlina, C. S., & Feinberg, J. M. (2025). Magnetic Characterization of Chondrites and their Suitability for Paleomagnetic Analysis of Refractory Inclusions. In *14th Institute for Rock Magnetism Conference*.
4. Sprain, C. J., Mittal, T., **Bristol, K.E.**, & Fendley, I. (2024). Quantitative Paleosecular Variation Analysis: A New Tool for Assessing Eruption Tempo of Large Igneous Provinces. In *AGU Fall Meeting Abstracts* (Abstract V21C–328).
5. **Bristol, K.E.**, Sprain, C. J., Mittal, T., Monteiro, A., Duraiswami, R. A., Tremblay, M. M., & Mijjium, M. (2024). Mantle Plumes and Geomagnetic Intensity Variations: Insights from the Deccan Traps. In *AGU Fall Meeting Abstracts* (Abstract GP41C–3549).
6. Sprain, C. J., Mittal, T., **Bristol, K.E.**, Duraiswami, R. A., Tremblay, M. M., Mijjium, M., & Monteiro, A. (2022). Quantitative Paleosecular Variation Analysis: A New Tool for Assessing Time Using Paleomagnetism. In *AGU Fall Meeting Abstracts* (Abstract GP46A–01).
7. Sprain, C., Hurst, E., Paterson, G., Bono, R., **Bristol, K.E.**, & Biggin, A. (2021). Constraining the Behavior of eps-Fe₂O₃ During Paleointensity Experimentation. In *AGU Fall Meeting Abstracts* (Abstract GP42A–04).
8. **Bristol, K.E.**, Sprain, C. J., & Meert, J. (2021). Absolute Paleointensity for Precambrian-Aged Dikes from India. In *Geological Society of America Abstracts* (Vol. 53, No. 6, p. 366752). <https://doi.org/10.1130/abs/2021AM-366752>
9. Sanmartin, K., Piispa, E. J., Mandon, C., Roverato, M., **Bristol, K.E.**, Smirnov, A. V., & Trindade, R. I. (2020). Rock Magnetism, Paleomagnetism and Paleointensity of Imbabura Volcano (ecuador)-Implications for the Spatiotemporal Growth Model. In *AGU Fall Meeting Abstracts* (Abstract GP003–09).
10. **Bristol, K.E.**, Smirnov, A. V., Piispa, E. J., Kosterov, A., Kulakov, E., & Ramirez Navas, M. R. (2019). Rock Magnetic Investigation of the Daule Ordinary Chondrite. In *AGU Fall Meeting Abstracts* (Abstract GP43B–0797).
11. Piispa, E. J., Larrea, P., Choez, K., **Bristol, K.E.**, Roverato, M., Smirnov, A. V., & Mandon, C. (2019). Paleo- and rock-magnetic record of the Imbabura volcanic units: Implications for the tectonomagmatic evolution of the volcano and for the Earth's magnetic field at equatorial latitudes. In *International Symposium on Andean Geodynamics Abstracts* (Vol. 9).
12. Piispa, E. J., Larrea, P., Choez, K., **Bristol, K.E.**, Roverato, M., Smirnov, A. V., & Mandon, C. (2019). Paleo- and rock-magnetic record of the Imbabura volcanic units: Implications for the

tectonomagmatic evolution of the volcano and for the Earth's magnetic field at equatorial latitudes. In *Latinmag Letters*.

13. Foucher, M. S., **Bristol, K.E.**, Piispa, E. J., & Smirnov, A. V. (2018). Absolute geomagnetic paleointensity from the 1.1 Ga Baraga-Marquette dike swarm (Michigan, USA) obtained using the Shaw and pseudo-Thellier methods. In *AGU Fall Meeting Abstracts*.
14. Foucher, M., Engel, E., **Bristol, K.E.**, & Smirnov, A. V. (2017). Evolution of large lava flows in rift setting: Paleomagnetic and rock magnetic insights into the Greenstone Flow. In *IAVECEI Scientific American Assembly*.
15. **Bristol, K.E.**, & Smirnov, A. V. (2017). Rock magnetic investigation of the carbonaceous chondrules from the Allende meteorite. In *Michigan Technological University Pavlis Honors College Undergraduate Research Symposium*.

HONORS AND AWARDS

I've been involved in grants/fellowships/awards totaling **\$397,280** as the principal recipient.

Awards:

- **Outstanding Student Presentation Award (OSPA)** – \$400 2023
Given for the most exceptional presentations during the annual 2023 AGU meeting.
- **Dr. Nancy Scofield Pioneering Research Award** – \$2,000 2019
Given to a graduate student whose work expands the boundaries of research in the GMES Dept. at Michigan Tech. University. I was the first undergraduate to receive this honor.
- **Field Geophysics Award** – \$50 2019
Given for exceptional performance in the field course Field Geophysics (GE3900) at MTU.
- **Departmental Scholar** – \$200 2017
Given to a student that best represents scholarship in the GMES Dept. at MTU.
- **Mich. Tech Presidential Scholars Award** – \$20,250 2014
Given to academically talented first-year undergraduates.
- **Mich. Tech Summer Youth Scholars Award** – \$13,500 2014

Fellowships:

- **NSF Astronomy & Astrophysics Postdoc Fellowship** – \$330,000 2025
- **UF CLAS Dissertation Fellowship** – \$12,780 2024
- **UMN Institute of Rock Magnetism Visiting Fellowship Award** – \$4,750 2022
- **NASA Michigan Space Grant Fellowship** – \$2,500 2018
- **MTU Summer Undergraduate Research Fellowship** – \$4,000 2016

Grants:

- **AGU Student Travel Grant** – \$1,000 2023
- **UF Geological Sciences Dept. Travel Grants** – \$2,000 2021, 2022, 2023
- **UF CLAS Grad Student Travel Grants** – \$600 2021, 2022
- **GSA Grad Student Research + Travel Grant** – \$3,000 2022
- **MTU Grad Student Gov. Travel Grant** – \$250 2019

PEER REVIEW AND SERVICE

- Geochemistry, Geophysics, Geosystems, Reviewer
- Geological Society of America Bulletin, Reviewer
- NASA Solar System Workings, Executive Secretary
- Journal of African Earth Sciences, Reviewer

TEACHING EXPERIENCE

University of Minnesota

IRM Summer School for Rock Magnetism: Instructor, 30 students

Summer 2024

University of Florida

GLY 3603C Paleontology: TA, 60 students

Spring 2024

GLY 3163 Geology of American National Parks: TA, 150 students

Spring 2023

Michigan Technological University

GE 3040 Fundamentals of Geophysics: TA, 11 students

Spring 2020

GE 4530 & GE 5430 Planetary Geology/Geophysics: Instructor, 8 students

Fall 2019

OUTREACH AND MENTORING

Outreach Events/Visits:

EM = Elementary/Middle School, HS = High School, UG = Undergraduate

- **Physics Inside Out**, Purdue University 2025
Instructor, (~ 24 EM students annually)
- **Girls Do Science**, Florida Museum of Natural History 2023 - 2024
Exhibitor, (~1000 Children + EM/HS students attend annually)
- **Scientist in Every Florida School**, Thompson Earth Systems Institute 2022 - 2024
Visiting Scientist, Collier County Public Schools (~240 EM students)
- **Can You Dig It?**, Florida Museum of Natural History 2022 - 2024
Exhibitor, (~1500-2000 EM+/HS students attend annually)
- **Summer Youth Programs**, Michigan Tech. University 2015 - 2019
Instructor, curriculum developer, and field trip advisor for:
 - Geological Engineering Exploration Program (~150 EM + HS students)
 - Women in Engineering Program (~450 EM + HS students)
 - Engineering Scholars Program (~420 EM + HS students)

Mentor:

- **Assistant Research Mentor**, University of Florida 2020 - 2024
5 UG students
- **Assistant Research Mentor**, Michigan Tech. University 2015 - 2020
1 HS, 2 UG students, & 2 HS STEM teachers (NSF K-12 Outreach)

SKILLS

General	Research, Technical Writing, Data Analysis, Public Speaking Science Communication, Curriculum Development, Grant Writing, and Laboratory Management
Programming & Software	Python, R, MATLAB, LaTeX, Mathematica, SQL, Git, Software Development, and Machine Learning
Instrumentation	Magnetometry, Paleomagnetic Furnaces, Cryogenics, Mineral Separators, Scanning Electron Microscopy, and Quantum Diamond Microscopy

FIELD EXPERIENCE

Deccan Traps, Maharashtra, India	– Sample collection	3 weeks, 2022
Custer National Forest, Montana, USA	– Sample collection	1 week, 2021
Hell Creek Region, Montana, USA	– Sample collection	1 week, 2021
Calumet, Michigan, USA	– Geophysical surveying for an archaeological excavation	2 days, 2018
Keweenaw Peninsula, Michigan, USA	– Geophysical surveying	5 weeks, 2018
Midcontinent Rift, Michigan, USA	– Sample collection	Various dates, 2015 - 2018
North and South Islands, New Zealand	– Sample collection/mapping	6 weeks, 2017